

Research Redesign Plan for an Acceptable ATM-Crime Paper

Executive recommendation

After reading the current manuscript and all three sets of reviews, my strongest recommendation is **not** to spend your remaining time inventing a more complicated model and **not** to turn the paper into a descriptive NYC Open Data study.

The most defensible redesign is:

Build a multi-city, exposure-aware ATM-site crime model and ask whether environmental risk factors add predictive information beyond the surrounding neighborhood's ordinary crime level—and whether those relationships transfer from one city to another.

A working title could be:

Beyond Crime Hotspots: Do Environmental Risk Factors for ATM-Site Crime Transfer Across Cities?

or, if you stay entirely within New York State:

Do ATM Crime-Risk Models Travel? Background-Adjusted External Validation Across New York Cities

The core research questions would be:

RQ1. After controlling for the surrounding area's baseline crime level, do characteristics of an ATM's immediate environment still explain or predict ATM-site crime?

RQ2. Which environmental risk factors are stable across cities, and which are city-specific?

RQ3. How much predictive performance is lost when a model trained in one city is transferred to another city?

RQ4. Can a small amount of target-city data recover that lost performance?

This directly solves your first concern. You are no longer reporting the unsurprising result that "high-crime places have more ATM crime." **Background crime becomes the baseline that your proposed method must beat.**

It also gives you a realistic version of your second idea. You do **not** need to discover every causal factor behind ATM attacks, and you do **not** need to build a sophisticated deep domain-adversarial network. Published cross-city crime research already demonstrates that common urban features can be harmonized and transferred across cities using substantially simpler ensemble/instance-transfer designs. Bappee et al., for example, transferred crime-prediction information from Toronto and Vancouver to Halifax using common feature representations and Gradient Boosting, while showing that including even some target-city data can improve the transfer. ¹ More advanced unsupervised domain-adaptation models also exist, confirming that cross-city crime transfer is an established research problem, but you do not need that level of complexity for this paper. ²

Most importantly, the novelty becomes **scientific rather than algorithmic**:

Are ATM-site risks simply reflections of generic neighborhood crime, or is there an ATM-specific environmental risk signal that remains predictive across urban environments?

That is a substantially stronger research question than "which ML model predicts ATM crime best?"

My ranking of your options is therefore:

Direction	Novelty	Difficulty	Addresses reviewers?	Recommendation
NYC Open Data + descriptive charts	Low	Low	Poorly	Do not make this the paper
Replace DT with a "better" ML model	Low	Medium	Poorly	Do not make this the contribution
Full deep-learning/domain-adaptation system	High	Very high	Partly	Too risky for your timeline
Lisbon-only repair	Moderate-low	Medium	Yes	Fallback option
Background-adjusted multi-city transfer study	High enough	Medium	Very strongly	Recommended

The last option is also directly supported by the literature. Cross-jurisdictional Risk Terrain Modeling has found that robbery-related environmental factors can be similar across cities without being identical, and that their spatial influence can differ by urban environment. That is almost exactly the empirical question your ATM-specific study could sharpen. ³ Recent work has likewise demonstrated the feasibility of using common OpenStreetMap/register features for spatial risk models across cities of different sizes. ⁴

What the reviews are actually telling you

The three rejection rounds look different on the surface, but they converge on a small number of fundamental problems. You should treat these as design requirements for the new paper.

The most important observation is that **the problem is not primarily that your current model is too simple**. A substantially more complicated model trained on the same target construction would still fail.

The current label problem is fatal unless you remove it

The PRICAI reviewers identified the most serious issue: only 107 of the 746 Lisbon ATMs have directly observed attack information, while the labels for the remaining 639—about 86%—were constructed using expert judgment together with geographical context and parish-level risk. The same geographical variables, particularly motorway and police-station distance, are then used by the model to "discover" important predictors. [filecite turn0file0](#)

That creates a genuine circularity problem:

geography → expert-generated label → model → "geography is important"

Your current PRICAI manuscript tries to strengthen the analysis with SEM, decision trees, SHAP, cross-validation, fuzzy inference, and Choquet aggregation, but the fundamental target problem remains. The manuscript says that 107 ATM records are directly observed, but its classification section subsequently describes 88 attacked and 658 non-attacked records across the full 746-site data and applies SMOTE before fitting the decision tree. [filecite turn0file3](#)

That means the methodological chain is sophisticated **after the point at which the validity problem has already entered the data**.

No Random Forest, XGBoost, neural network, transformer, GNN, or transfer-learning method can repair that.

The paper has accumulated methods rather than a single testable contribution

The current manuscript contains:

SEM → Decision Tree → SHAP → Fuzzy Inference → Choquet Integral → sensitivity analysis.

That looks sophisticated, but each additional component creates another question for reviewers:

- Why SEM?
- Why not a multilevel model?
- Why decision trees?
- Why only Mway and Pdis?
- Why fuzzy inference?
- How were memberships selected?
- Why Choquet?
- Why are the Choquet inputs independent?
- Why does Choquet give almost the same ranking as a weighted mean?
- What is actually being predicted?
- Is this prediction, explanation, or multi-criteria decision support?

The Security Journal reviewer explicitly suggested that the contribution did **not** have to come from claiming methodological novelty; a useful dataset, sound empirical insight, and appropriate established methods could form a better contribution. [filecite\turn0file1](#) The IEEE BigData rejection similarly criticized the work as an application of classical tools to one old, geographically specific dataset without enough evidence that the result generalizes. [filecite\turn0file2](#)

This is actually good news: **you do not need a new algorithm to get this paper accepted.**

You need a much cleaner experiment.

The Choquet component should probably be removed

In your current manuscript, the Choquet ranking has a Spearman correlation of **0.9985** with the additive alternative, and the top-75 ATM set is identical. The manuscript now explains why that happens mathematically, but explaining why a complex method is almost equivalent to a simple method does not demonstrate why readers should use the complex method. [filecite\turn0file3](#)

The reviewers noticed exactly that. [filecite\turn0file0](#)

There is a second problem. Your security criterion c_1 is a nonlinear function of motorway/police distance while your countermeasure criterion c_3 is again constructed from motorway/police distance. Even if their Pearson correlation is reduced by the nonlinear FIS transformation, they are still conceptually derived from overlapping information. [filecite\turn0file3](#)

My recommendation is therefore:

Drop the Choquet integral from the main new paper.

If you eventually need a placement decision layer, risk, accessibility, and cost can be presented as separate objectives, potentially through a Pareto frontier. A transparent weighted ranking can be included only as a practical illustration with sensitivity analysis. You should not make the aggregation mechanism the claimed scientific contribution.

The "greenfield placement" claim is stronger than the experiment supports

The present study claims to solve greenfield ATM placement, but almost all empirical validation concerns the ranking of **existing** ATM locations. [filecite\turn0file3](#)

A genuine greenfield study needs a set of candidate locations at which no ATM currently exists, together with demand/accessibility and security features, followed by a defensible evaluation of recommendations.

If you do not have that, I would change the language from:

"optimal greenfield ATM placement"

to:

"ATM-site crime-risk assessment" or "security-aware ATM site evaluation."

You can still explain in the discussion how such a risk estimator could become an input to future ATM placement. That is much safer than claiming an optimization problem you have not actually validated.

Validation must move from aggregated correlation to external prediction

The existing parish-level Spearman result is interesting, but it averages site-level scores over only 21 parishes. The PRICAI reviewers correctly pointed out that this does not establish whether individual risky ATMs are ranked correctly. [filecite turn0file0](#)

You need a paper in which the strongest figure is something like:

Train in NYC → test in Buffalo.

Train in NYC + Buffalo → test in Rochester.

Train on 2023–2024 → predict 2025.

That is far more convincing evidence of generalization than a correlation on the same geographical system used to construct the model.

This is also methodologically important because observations close together in space or time are not independent. Random cross-validation can substantially underestimate predictive error when such dependence is ignored. [5](#)

The research question I would build the new paper around

The strongest version of your idea is not simply "transfer learning for ATM attacks." I would formulate it as:

How transportable is ATM-site crime risk across urban environments after accounting for background neighborhood crime?

That formulation gives you three layers of novelty.

First, it separates **ATM-specific environmental risk** from ordinary neighborhood crime.

Second, it tests **external validity across cities**, addressing the repeated criticism that Lisbon-specific findings may not generalize.

Third, it gives transfer learning a clear purpose: **can a bank estimate risk in a data-poor city using data from a data-rich city?**

Why "background-adjusted risk" solves your NYC novelty problem

Suppose an ATM is in an area where crime is already extremely high. Finding many incidents around that ATM teaches us very little.

So make the naïve model your baseline:

$$M_0 : Y_{\text{ATM}} = f(\text{background neighborhood crime})$$

Then test whether adding the ATM's environmental context improves out-of-sample prediction:

$$M_1 : Y_{\text{ATM}} = f(\text{background crime, road access, transit, land use, commercial activity, guardianship, urban density})$$

Your central empirical result becomes:

$$\Delta = \text{Performance}(M_1) - \text{Performance}(M_0)$$

If Δ is essentially zero, that is still a meaningful result: ATM-site risk mostly reflects general neighborhood crime.

If $\Delta > 0$ and replicates across cities, you have much stronger evidence that there is an ATM-specific environmental signal.

If $\Delta > 0$ within cities but disappears during city-to-city transfer, you have another interesting finding: ATM risk is **context dependent and requires local calibration**.

All three outcomes are scientifically informative.

That is fundamentally different from searching for whichever features maximize accuracy.

The transfer-learning question can be made manageable

"Transfer learning" sounds like you need to identify every variable driving crime and learn a universal representation. You do not.

For your project, **transfer means holding the feature definition constant while changing the geographical domain**.

For example:

Source domain = *NYC*

Target domain = *Buffalo*

Both cities contain ATM sites represented by the same variables:

$X = [\text{background crime, road proximity, intersection density, transit accessibility, population density, commercial}]$

and approximately the same outcome definition:

$$Y = \text{robbery/burglary/larceny event near an ATM}$$

You train:

$$f_{\text{NYC}}(X)$$

and evaluate it directly on:

$$D_{\text{Buffalo}}.$$

That alone is a **zero-shot cross-city transfer experiment**.

You can subsequently provide the model a small fraction of Buffalo labels and measure how much performance recovers. That is the manageable transfer-learning component.

This has precedent. Bappee et al. explicitly treated Toronto/Vancouver and Halifax as related but differently distributed urban domains, used common crime/demographic/POI variables, evaluated source-only and source-plus-target configurations, and found that incorporating target-specific observations was important.

¹ Other work on cross-city crime risk has gone further with unsupervised domain adaptation, demonstrating that context inconsistency between cities is itself an important modeling challenge. ²

Your contribution does not need to reproduce the complicated latter model. Your contribution can be the **ATM-specific test of transportability**.

The three-city New York design is particularly attractive

I recommend beginning with:

New York City + Buffalo + Rochester.

There is a very practical reason.

New York State's Department of Financial Services publishes a statewide dataset of individual **bank-owned or in-bank ATM locations**, with statewide coverage and a monthly posting schedule. The current data page explicitly identifies individual ATM location as the granularity. ⁶

The limitation is important: it does **not** contain independent non-bank ATMs such as machines in delis or bodegas. ⁶

Rather than treating that as fatal, define your population precisely:

Bank-owned or in-bank ATM sites in New York State.

That gives NYC, Buffalo, and Rochester a single, administratively consistent source for the ATM denominator.

For crimes, NYC's official NYPD Historic Complaint dataset provides geocoded reported felony, misdemeanor, and violation complaints. ⁷ Rochester's police department provides Part I crime data including robbery, aggravated assault, burglary, larceny, homicide, and motor-vehicle theft, with data available from 2011 and new data added daily. ⁸ Buffalo's official open-data portal likewise publishes Crime Incidents data. ⁹

The result is much cleaner than attempting Lisbon → NYC transfer, where differences in labels, exposure periods, data-generation procedures, and ATM definitions would make it difficult to tell whether poor transfer reflected genuine geography or incompatible measurement.

Data design: this is where the new paper will succeed or fail

This is more important than choosing between Random Forest and XGBoost.

Your uploaded NYC CSV is useful, but it is not yet a modeling dataset

I inspected `sorted_data.csv`. It contains **7,537 rows and 35 columns**, and every row has `PREM_TYP_DESC = ATM`.

But these 7,537 rows are **crime/complaint events**, not 7,537 ATM machines.

That distinction is crucial.

The data contain:

- 3,787 Grand Larceny records;
- 1,535 Petit Larceny records;
- 462 Robbery records;
- 423 Theft-Fraud records;
- 415 Criminal Mischief-related records;
- 282 Forgery records;
- 108 Burglary records.

So roughly 71% of rows are Grand or Petit Larceny alone, while other records represent robbery, fraud, forgery, criminal mischief, and other mechanisms.

There are only around 4,400 distinct rounded coordinate locations despite 7,537 complaint rows, meaning repeated events at the same or nearby locations are common.

This tells us two important things.

First:

Do not turn each complaint into an independent ATM observation.

One ATM can have multiple complaints.

Second:

Do not call all 7,537 records "physical ATM attacks."

ATM-premise larceny, ATM transaction fraud, card-related theft, robbery of an ATM user, physical machine damage, and burglary are criminologically different processes. The official NYPD data are incident/complaint records, not a dedicated dataset of physical attacks on machines. ⁷

This is exactly the kind of construct-validity ambiguity that hurt the Lisbon paper.

Freeze one primary outcome before doing any modeling

I recommend that the main paper study something like:

ATM-site acquisitive crime risk

rather than:

ATM attack probability.

A conservative cross-city definition could use harmonizable categories such as:

$$Y = \{\text{robbery, burglary, larceny/theft}\}$$

occurring at, or within a predefined radius of, a bank-owned ATM.

However, because some NYC "larceny" descriptions involve bank-account compromise rather than a physical event, you should build a codebook from `OFNS_DESC`, `PD_DESC`, and offense codes and explicitly exclude purely transactional/financial fraud from the physical-site primary outcome.

One possibility is to define:

Outcome	Examples	Status
Primary: physical/acquisitive ATM-site crime	robbery, burglary, appropriate theft categories	Main paper
Property damage	criminal mischief, machine damage where identifiable	Secondary analysis
Financial/transaction fraud	bank-account compromise, forgery, skimming-related records	Separate outcome or excluded

Do **not** merge these simply to increase sample size.

Mechanistically, motorway access might matter for a physical attack but be irrelevant to electronic account compromise. Combining both and asking SHAP which feature is important produces difficult-to-interpret averages.

Create the ATM denominator first

Your modeling table should begin from the full list of ATM sites:

atm_id	city	latitude	longitude	bank	year	...
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Then crime events are linked to those sites.

This gives you both:

- ATM sites with observed incidents;
- ATM sites with no observed incidents during the observation window.

That fixes one of the fundamental issues in the old Lisbon setup.

The New York State ATM dataset is well suited to that role because it is explicitly an individual-location dataset of bank-owned/in-bank machines. ⁶

There is, however, an important temporal problem: the dataset describes the **current** ATM universe and is updated monthly. ⁶

Therefore:

Do not join a 2026 ATM snapshot to twenty years of 2006–2025 crime and assume every current ATM existed throughout that period.

That would introduce another exposure bias.

Your first data-engineering task should be to determine whether historical monthly/yearly ATM snapshots can be recovered. If they can, construct ATM-years.

If historical locations cannot be recovered, restrict the main analysis to a recent period for which the current ATM universe is reasonably defensible, explicitly describe the assumption, and run a sensitivity analysis.

This is far better than quietly treating unequal time-at-risk as if it did not exist—the exact issue raised by the Security Journal reviewer. ⁶filecite⁶turn0file1⁶

Recommended unit of analysis

The best unit is:

ATM site $\times$ time period

For example:

ATM i during calendar year t

or, if data are sparse:

ATM i during a two-year observation window.

Then define either a count:

Y_{it} = number of target incidents within 50 m of ATM i

or a binary endpoint:

$$Y_{it} = \begin{cases} 1 & \text{if at least one target incident occurs} \\ 0 & \text{otherwise.} \end{cases}$$

I prefer the **count formulation** if enough sites have repeated incidents, because it preserves more information.

For spatial linkage, pre-register a primary radius, perhaps 50 m, but repeat the analysis at 25 m and 100 m. The purpose is not to search for whichever radius gives the best result; it is to show that conclusions are not artifacts of one arbitrary buffer.

Make background crime an explicit baseline

This is the most important design innovation for answering your first concern.

For every ATM i , calculate previous-period crime in the surrounding neighborhood while **excluding the immediate ATM buffer**.

For example:

B_{it} = target crimes during $t - 1$ within 200–500 m of ATM i

The inner exclusion prevents the thing you are trying to predict from simply being counted in the predictor.

Then construct:

Baseline model

$$M_0 : Y_{it} \sim B_{i,t-1}$$

Environmental model

$$M_1 : Y_{it} \sim B_{i,t-1} + X_i$$

where X_i contains site/environmental attributes.

Now a reviewer can no longer say:

"Of course there are more ATM crimes where there is more crime."

Your response is:

"Yes. That is our explicit baseline. The research question is whether ATM-site context contains **incremental out-of-sample information above that baseline**, and whether that incremental information transfers across cities."

That is a clean research contribution.

Use common features rather than trying to find every conceivable cause of crime

You were worried that transfer learning requires discovering every variable that causes an ATM attack. It does not.

You need a **small, theoretically defensible common feature schema** available for all target cities.

A strong core would be:

Feature family	Examples	Why include it
Ambient risk	lagged surrounding robbery/burglary/larceny	Must-beat baseline
Road accessibility	distance to major road, road density, intersection density	accessibility/escape opportunity
Guardianship	police-station distance or similar proxy	guardianship context
Activity/accessibility	transit access, employment density, destination accessibility	human activity
Land use	commercial intensity, mixed use	target/activity concentration
Urban form	population density, walkability/street design	environmental context
ATM competition/exposure	nearby ATM density, nearest ATM distance	site availability/concentration
Socioeconomic context	income/unemployment at area level	contextual control

The EPA Smart Location Database is particularly useful here because it provides a nationwide block-group resource with more than 90 attributes covering density, land-use diversity, neighborhood design, destination accessibility, transit, employment, and demographics. ¹⁰

The ACS provides social, economic, housing, and demographic estimates down to the census block-group level. ¹¹ TIGER/Line geography can supply consistent boundaries/road-related spatial layers and contains codes that can be linked to Census demographic data. ¹²

This gives you something the Lisbon paper did not have:

the same variable definitions across all study cities.

Recent spatial-crime research demonstrates that common OpenStreetMap/place-feature datasets can support models across multiple cities, reinforcing the feasibility of this approach. ⁴

I would keep race/ethnicity out of the operational risk model unless you have a very specific fairness-analysis rationale. The model is intended to support access to financial services; using protected demographic characteristics to make neighborhoods appear "undesirable" raises ethical and interpretation problems that are unnecessary for your main research question.

Modeling and transfer-learning experiments

Your third idea—"find a better model"—should become a **model comparison inside the study**, not the paper's novelty claim.

A good model here is not simply the one with the highest training accuracy.

It should:

1. generalize geographically and temporally;
2. handle rare incidents;
3. represent nonlinear relationships where necessary;
4. produce reasonably calibrated probabilities or risks;
5. remain interpretable enough to explain environmental relationships;
6. work with your modest tabular dataset.

You therefore need only a few models.

A simple statistical baseline

If your outcome is a count, start with a **Negative Binomial regression**:

$$\log E[Y_{it}] = \alpha_{\text{city}} + \delta_t + \beta_0 B_{i,t-1} + \sum_j \beta_j X_{ij}.$$

Why this is useful:

- reviewers understand it;
- coefficients are interpretable;
- it gives you statistical uncertainty;
- it provides a serious benchmark against which ML must improve.

If the outcome is binary, use regularized logistic regression instead.

This model alone would already be methodologically cleaner than the current SEM → DT → FIS chain.

One strong nonlinear tabular model

Use **XGBoost** or **LightGBM** as the main flexible predictive model.

Not because it is "state of the art," and not because using XGBoost is novel.

Use it because your likely relationships are nonlinear—for example, the effect of road distance may change rapidly near an ATM and flatten after a few hundred meters—and because tree boosting can capture interactions without requiring you to specify each one manually.

Explain it exactly that way.

Do not write:

"We propose a novel state-of-the-art XGBoost framework."

Write:

"Gradient-boosted trees were selected as a flexible nonlinear benchmark for tabular spatial covariates; the scientific contribution lies in the exposure-adjusted and cross-city evaluation rather than the learning algorithm."

That wording directly avoids one of the prior reviewers' complaints about presenting established algorithms as methodological novelty. fileciteturn0file1

Optional third benchmark

You may include Random Forest because it is familiar and useful as a standard comparison.

You do not need ten algorithms.

A paper with:

- Negative Binomial/logistic regression;
- Random Forest;
- XGBoost/LightGBM;

is sufficient.

In spatial-crime research, more flexible machine-learning approaches have already been used to model micro-place risk, while recent literature increasingly combines prediction with interpretability rather than treating model complexity itself as the contribution. ¹³

Explainability should come after valid prediction

SHAP can remain, but it should answer:

"What drives an externally validated model?"

rather than:

"Which feature looks important on the same data used to construct the outcome?"

So first show that the model predicts held-out data.

Then show:

- SHAP summary;
- partial dependence or accumulated local effects;
- feature ranking;
- perhaps city-specific effect plots.

The most interesting figure might be:

Feature	NYC	Buffalo	Rochester	Stable?
Background crime	strong +	strong +	strong +	Yes
Major-road access	+	+	weak	Partial
Commercial activity	+	+	+	Yes
Police distance	mixed	weak	mixed	No
Transit accessibility	+	weak	+	Partial

That would be much more meaningful than simply saying "Pdis had SHAP = 0.17."

It also gives you a way to revisit the Lisbon result. If motorway proximity transfers but police-distance direction does not, you can argue that the earlier police result was probably context-specific or a proxy for urban centrality—exactly the confounding possibility raised by the Security Journal reviewer.

filecite turn0file1

Your transfer experiment should have a ladder of increasing target information

For each target city T , run the following.

Experiment	Training data	Testing data	Question
Local-full	Earlier data from T	later data from T	How well can we do locally?

Experiment	Training data	Testing data	Question
Zero-shot	Source city only	T	Does the source model travel at all?
Multi-source	Other two cities	T	Does multi-city training travel better?
Few-shot transfer	Other cities + 10% of T	remaining T	Does little local data help?
More-target transfer	Other cities + 25% of T	remaining T	How much local data is enough?
Target-small baseline	Same 10% or 25% of T , no source	remaining T	Is transferred data actually helping?

That last comparison is essential.

Suppose:

$$AUPRC(\text{source} + 10\% \text{ target}) = 0.34$$

and

$$AUPRC(10\% \text{ target only}) = 0.22.$$

Then the source-city data genuinely helped.

But if both are 0.34, your "transfer learning" did nothing; the target data alone explains the improvement.

Define two simple quantities:

$$\text{Transfer Gap} = P_{\text{local-full}} - P_{\text{zero-shot}}$$

and

$$\text{Transfer Benefit}(k) = P_{\text{source}+k\% \text{ target}} - P_{k\% \text{ target only}}.$$

These two plots could become central results of the paper.

This design is directly motivated by prior cross-domain crime work: Bappee et al. found that source-only transfer performed worse than configurations incorporating target-city observations and explicitly examined source/target feature-distribution differences. ¹

You can make the research question even more interesting by asking:

How much local labeled data is needed before a transferred ATM-risk model becomes useful?

That is practical for banks entering a city where they have little historical incident information.

You do not need deep domain adaptation in version one

There is published work using a dense convolutional/domain-adaptation architecture to construct domain-invariant representations for cross-city crime risk. ²

I would cite that work but deliberately position your project differently:

"Rather than proposing a high-capacity domain adaptation architecture, we test whether a parsimonious common feature representation and limited target calibration are sufficient for ATM-specific risk transportability."

That turns "we are not sophisticated enough to implement DANN" into a methodological strength:

simple, auditable, deployable transfer.

If your baseline paper works and you later have time, TrAdaBoost or instance reweighting could be added as one transfer baseline because prior crime research has used that approach. ¹

But do the source-only, pooled, and few-shot experiments first. Those are enough to answer the main scientific question.

Validation and the experiments reviewers will expect

The redesign will only work if the evaluation is much stricter than the current paper.

Separate time, space, and cities

Your strongest hierarchy is:

random split < spatial holdout < future-time holdout < new-city holdout.

The exact validation strategy should reflect the intended deployment. Since your claim is that the method can assess unfamiliar sites/cities, nearby observations should not freely leak between training and test sets.

At minimum:

Within-city temporal validation

train previous period → test later period.

Within-city spatial robustness

Hold out spatial groups such as precincts, neighborhoods, or sufficiently large grid blocks.

Across-city external validation

$$NYC \rightarrow Buffalo, \quad NYC \rightarrow Rochester, \quad Buffalo + Rochester \rightarrow NYC$$

and rotate the city held out.

This directly tests the generalization criticism from the IEEE BigData review. [filecite turn0file2](#)

Structured cross-validation matters because ignoring spatial or temporal dependencies can lead to serious underestimation of predictive error. [5](#)

Put preprocessing inside the training folds

This is particularly important because the current manuscript describes SMOTE balancing followed by cross-validation in a way that is ambiguous about ordering. [filecite turn0file3](#)

In the redesigned analysis, every learned transformation must be fitted **only on the current training fold**, including:

- imputation;
- scaling;
- feature selection;
- hyperparameter tuning;
- any over/under-sampling;
- calibration.

Never construct a balanced dataset once and then divide that already-resampled dataset into train/test folds.

In fact, with XGBoost/LightGBM or weighted logistic regression, you may not need SMOTE at all. Class weights and an appropriate loss can be cleaner.

Accuracy should not be your main metric

ATM incidents will likely be rare relative to non-incident ATM-site periods.

Imagine that only 5% of ATM-years have an event.

A classifier that predicts "safe" for every ATM gets:

95% accuracy

while being useless.

For a binary endpoint I would report:

Primary

AUPRC

because Precision-Recall evaluation is particularly informative for imbalanced binary outcomes. ¹⁴

Also report

- ROC-AUC;
- Brier score;
- calibration plot;
- recall in the highest-risk 5% and 10% of sites;
- precision in the highest-risk 5% and 10%;
- lift over random targeting.

For example:

$$\text{Top-10\% Lift} = \frac{\text{fraction of future incidents captured in top 10\% risk sites}}{0.10}.$$

If 37% of incidents fall in your highest-risk 10% of ATMs:

$$\text{Lift} = 3.7.$$

That means something operationally meaningful to a bank.

Spatial crime research routinely evaluates whether a relatively small high-risk area captures a disproportionate share of future events; Risk Terrain Modeling research is built around this concept, and systematic evidence supports the value of evaluating concentrated high-risk areas rather than relying only on generic classification statistics. ¹⁵

For count outcomes, add Negative Binomial/Poisson deviance, MAE, and ranking metrics.

Include uncertainty, not only point estimates

Report bootstrap confidence intervals for:

- AUPRC;
- top-k recall;
- calibration;
- transfer gap;
- transfer benefit.

For example:

$$\Delta AUPRC = 0.071, 95\% CI = [0.031, 0.112].$$

That is much stronger than writing "XGBoost performed better than Logistic Regression."

For model comparisons, use paired bootstrap resampling so both models are evaluated on the same held-out observations.

Pre-register the important sensitivity checks

At minimum run:

$$r \in \{25, 50, 100\} \text{ meters}$$

for incident-to-ATM matching.

Also test:

- narrow vs broader crime definitions;
- one-year vs two-year outcome windows;
- with vs without background-crime predictor;
- with vs without demographics;
- bank-owned ATMs only, clearly acknowledging the missing independent ATM population;
- absolute variables vs standardized/common city representation.

The state ATM dataset's omission of non-bank machines means the conclusions cannot be generalized uncritically to every ATM in New York. ⁶ State that directly in the limitations rather than allowing a reviewer to discover it.

Add a negative-control style baseline

This is one of the best ways to make the paper feel scientifically serious.

Compare your environmental model against:

$$M_{\text{background}} = \text{lagged surrounding crime only.}$$

Then compare:

$$M_{\text{full}} = \text{background crime} + \text{ATM environment.}$$

A further baseline could use:

$$M_{\text{history}} = \text{previous ATM-site incidents only.}$$

Your model is valuable only if it beats one or both.

That prevents you from "winning" merely because crime repeats at the same places.

Recent cross-city spatial-risk research explicitly compares models with and without prior crime, providing a useful precedent for separating environmental risk information from crime-history information. ⁴

How I would restructure the paper itself

The new manuscript should feel like **one research question answered rigorously**, rather than several algorithms connected in a pipeline.

A clean structure would be:

Paper section	What it should do
Introduction	Explain ATM-site risk, the "generic hotspot" problem, and transportability gap
Related work	ATM crime/security; crime-and-place/risk terrain; cross-city prediction/domain adaptation
Data	Define ATM universe, incident taxonomy, exposure window, common covariates
Problem formulation	Formalize incremental risk and cross-city transfer
Methods	Three models maximum + transfer experiments
Validation	Temporal, spatial, and city holdouts; metrics; leakage controls
Results	Baseline value → environmental lift → city transfer → few-shot recovery
Interpretation	Stable vs city-specific predictors
Discussion	Operational implications, bias, limits, bank-owned-only coverage
Conclusion	What actually transfers and how much local data is needed

Your introduction's central gap could read approximately:

Existing spatial crime models establish that environmental characteristics influence micro-place crime risk, but cross-jurisdictional studies show that the relevance and spatial influence of those characteristics need not remain constant between cities. ³ Cross-domain crime-prediction studies further demonstrate that models can exploit information from other urban areas but can suffer when source and target feature distributions differ. ¹⁶ Yet the transportability of environmental risk factors for ATM sites, particularly after controlling for the surrounding background crime environment, remains largely untested. We therefore ask whether ATM-site risk contains a transferable place-specific component beyond generic neighborhood crime.

That is a much clearer paper than:

"We integrate SEM, CART, FIS, SHAP and the Choquet integral."

Your contributions could then be stated as:

Contribution A — Data. A reproducible ATM-site dataset harmonizing bank-owned ATM locations, police incident records, and nationally consistent built-environment variables across multiple cities.

Contribution B — Scientific formulation. A background-adjusted evaluation determining whether ATM environmental features predict crime above ambient neighborhood risk.

Contribution C — Generalization. A city-to-city external-validation experiment measuring which ATM-risk relationships transfer and which are location-specific.

Contribution D — Practical transfer. A controlled experiment measuring how much target-city data is required for useful adaptation.

Notice that **none of those contributions says "we use XGBoost."**

That is exactly what you want.

What to do with Lisbon

I would not throw the Lisbon work away. I would change its role.

The old paper can motivate the new study:

A prior Lisbon analysis suggested motorway accessibility and police-station distance as important geographical indicators, but the single-city dataset and partial expert-derived labels prevented a strong test of transportability.

Then ask whether similar environmental relationships emerge independently in New York cities.

However, **do not mix the 639 inferred Lisbon labels into the new supervised model.**

If the Lisbon data are retained quantitatively, use only independently observed information.

And there is one critical question you must resolve before doing that:

Are the 107 "directly observed" ATM records a sample containing known attacked **and known non-attacked/exposed controls**, or are they primarily ATMs for which some incident information happened to be available?

If they are merely incident-observed records rather than a proper risk set, then even those 107 are not automatically a valid classification dataset.

For a valid binary study, you need:

$$P(\text{attack} \mid \text{ATM exposed})$$

which requires defensible attacked and non-attacked ATM exposure—not only information about where attacks happened.

The Security Journal reviewer already raised the closely related issue of unequal time-at-risk and the ambiguity surrounding multiple attacks at the same ATM. [filecite00turn0file1](#)

If you cannot reconstruct that exposure correctly, keep Lisbon as motivating/contextual evidence rather than forcing it into the new predictive experiment.

A concrete execution plan that fits a limited research timeline

I would run the project in stages so that you have a publishable fallback even if the full three-city experiment becomes difficult.

First phase: freeze the research construct

Before writing more code, produce a one-page specification containing:

Item	Decision to freeze
Population	Bank-owned/in-bank ATMs
Cities	NYC, Buffalo, Rochester
Primary outcome	One precisely defined acquisitive/physical-site crime category
Unit	ATM × year/window
Primary radius	e.g. 50 m
Sensitivity radii	25 m, 100 m
Background crime	Lagged incidents in outer surrounding area
Core covariates	10–20 common variables only
Primary metric	AUPRC for binary; deviance + ranking for counts
Main baseline	Background crime only
Main flexible model	XGBoost/LightGBM
Main statistical model	Logistic or Negative Binomial
External test	Leave-one-city-out

Do **not** change this specification after looking at which settings give nicer results unless the change is justified and reported as exploratory.

Second phase: build the ATM-site table

Start from the New York State ATM universe, not from the complaint dataset. The state dataset provides individual bank-owned/in-bank ATM locations and is updated monthly. ⁶

Then:

1. create unique ATM IDs;

2. remove or resolve duplicates;
3. attach city;
4. assign coordinates;
5. resolve historical exposure if available;
6. map crime events to the defined buffer;
7. aggregate events per ATM-period.

Your current NYC `sorted_data.csv` becomes an **event input**, not the master table.

Before scaling anything, manually inspect perhaps 100 randomly sampled event-ATM matches. Record:

- coordinate distance;
- offense description;
- whether it genuinely satisfies your primary crime definition;
- whether multiple complaints are duplicates;
- whether the event appears to describe a customer, machine, or transaction.

This small QA exercise could save the whole paper.

Third phase: create the common city-independent feature layer

Use the same code in all cities.

For each ATM, compute only ex-ante variables—things a bank would know before an incident occurs.

Examples:

$\log(1 + \text{distance to major road})$

intersection density_{500m}

commercial intensity_{500m}

population density

destination accessibility

transit accessibility

ATM density_{500m}

lagged background crime_{200–500m}

The EPA Smart Location Database is attractive for common urban-form variables precisely because the same definitions are available across U.S. block groups. ¹⁰ ACS provides corresponding socioeconomic information at block-group resolution. ¹¹

Do not chase dozens of city-specific variables during the first iteration. A transfer experiment becomes difficult to interpret when NYC has 45 features, Buffalo has 25, and Rochester has 17.

Fourth phase: establish whether there is a research result before doing transfer learning

Fit:

M_0 = background crime only

M_1 = background + common environmental features, logistic/NB

M_2 = background + common environmental features, boosted trees

using strict held-out prediction.

Ask:

Does M_1 or M_2 outperform M_0 ?

This is your first go/no-go checkpoint.

If **no**, do not waste two weeks implementing fancy transfer learning. You may already have a publishable null result if it is consistent across cities:

"After accounting for local crime conditions, ATM-site environmental variables offer little incremental predictive value."

If **yes**, continue.

Fifth phase: perform city-transfer experiments

For each target city:

other city/cities $\rightarrow$ target city

Record:

- AUPRC;
- top-decile recall;
- calibration;
- transfer gap.

Then expose the model to 10% and 25% of target labels and calculate transfer benefit.

The final transfer result could look like this:

Target	Local-full	Zero-shot	Other cities pooled	+10% target	10% target only
NYC	.XX	.XX	.XX	.XX	.XX

Target	Local-full	Zero-shot	Other cities pooled	+10% target	10% target only
Buffalo	.XX	.XX	.XX	.XX	.XX
Rochester	.XX	.XX	.XX	.XX	.XX

One table like that is more valuable to the paper than five additional algorithms.

Sixth phase: interpret only what survived external validation

Now run explainability.

Ask:

Which variables remain important in more than one city?

Does their direction remain stable?

Does their relationship have approximately the same nonlinear shape?

Cross-jurisdictional crime research provides a useful expectation: environmental risk factors can exhibit meaningful similarities while still varying in influence across urban settings. ³

So a finding such as:

"Major-road accessibility is stable across all three cities, whereas police-station proximity reverses direction between cities"

would be much more scientifically interesting than simply reporting that one variable had the largest SHAP score.

It could also explain why the Lisbon model looked the way it did.

Final phase: only then decide whether to add placement

Once you have a trustworthy risk score $R(x)$, the simplest ATM decision problem is:

$$\text{minimize } R(x)$$

subject to:

$$\text{Accessibility}(x) \geq A_{\min}$$

and perhaps:

$$\text{Demand}(x) \geq D_{\min}.$$

This is much cleaner than converting everything into one Choquet score.

If you have three objectives—risk, accessibility, and cost—you can show a **Pareto frontier**, where a site is retained if no other candidate simultaneously has lower risk, better accessibility, and lower cost.

That makes trade-offs visible rather than hiding them inside expert-specified fuzzy capacities.

Most importantly, do not call this "greenfield optimization" until your candidate set includes locations that are **not merely existing ATMs**.

Final recommendation

Your paper has already gone through enough iterations that another "method patch" is unlikely to work.

The reviews collectively say:

- the application is important;
- reproducibility is a strength;
- the Lisbon dataset is unusual and potentially valuable; `filecite{turn0file0}`
- but the target construction, evaluation, baselines, and generalization evidence are not yet strong enough.

So I would **stop developing the current SEM-DT-FIS-Choquet architecture**.

The next paper should be simpler technically and stronger scientifically.

The central design should be:

ATM-site risk above background crime
+ common environmental variables
+ strict future/spatial validation
+ cross-city external testing
+ simple few-shot transfer

The paper's headline question is:

Does an ATM have predictable environmental vulnerability beyond simply being located in a high-crime neighborhood, and does that vulnerability generalize to another city?

That question is non-trivial. It directly answers your concern about NYC becoming "just charts." It turns transfer learning into a manageable sequence of experiments rather than an enormous deep-learning project. It removes the pressure to invent a novel classifier. And, most importantly, it attacks almost every major weakness that the previous reviewers identified.

For your timeline, I would set the scope hierarchy as:

Preferred paper: NYC + Buffalo + Rochester, background-adjusted risk, leave-one-city-out transfer, logistic/NB + boosted trees.

Minimum viable paper: two New York cities, the same background-adjusted design, zero-shot plus 10% target adaptation.

Fallback paper: NYC only, but with a defensible ATM denominator, narrow outcome definition, lagged background-crime baseline, temporal/spatial holdouts, and point-level validation.

Last-resort Lisbon repair: throw away all inferred labels for supervised modeling, verify the actual exposure/control status of the 107 directly observed cases, use a small interpretable statistical model, and abandon the FIS/Choquet complexity unless you acquire independent data that make those components genuinely necessary.

Of these, the **multi-city background-adjusted design is the one I would invest the research time in**. It changes the paper from *"we applied several analytical methods to ATM data"* into *"we tested whether ATM-specific environmental risk exists beyond general crime concentration and whether that knowledge transports across cities."* That is a considerably stronger basis for a new submission.

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